EE/CprE/SE 491 WEEKLY REPORT 8

August 28, 2025 - September 4, 2025

Group number: *16*

Project title: *Outdoor Roomba*

Client &/Advisor: *Prof. Diane Rover*

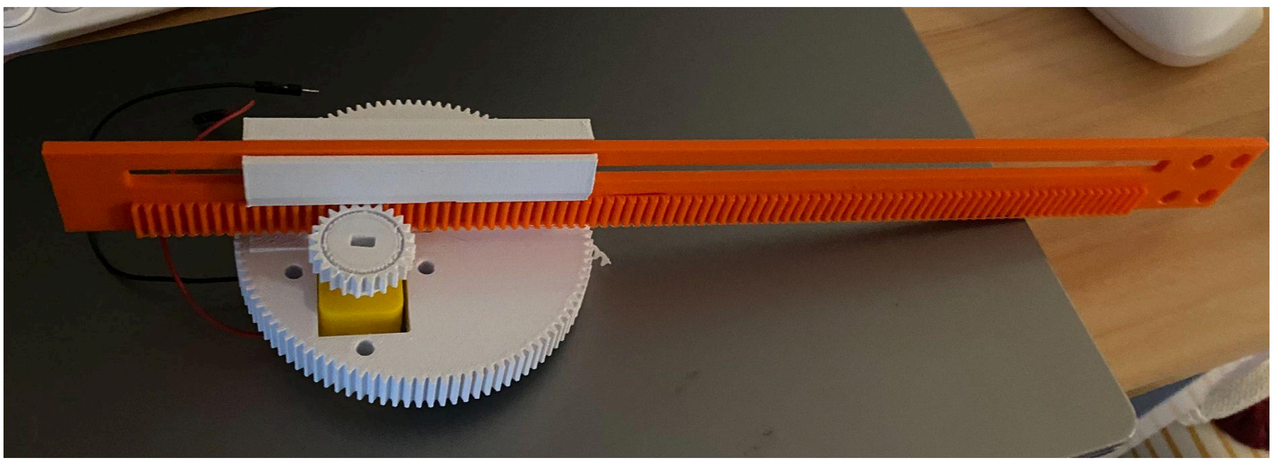
Team Members/Role:
Brodie M Bates - ***Embedded system***
Daniel Vergara Pinilla - ***Machine Learning***
Mitchell D Owen - ***Project Tester***
Om Shukla - ***Frontend Developer***
Sudipta Halder - ***Simulation developer***

- **Weekly Summary**

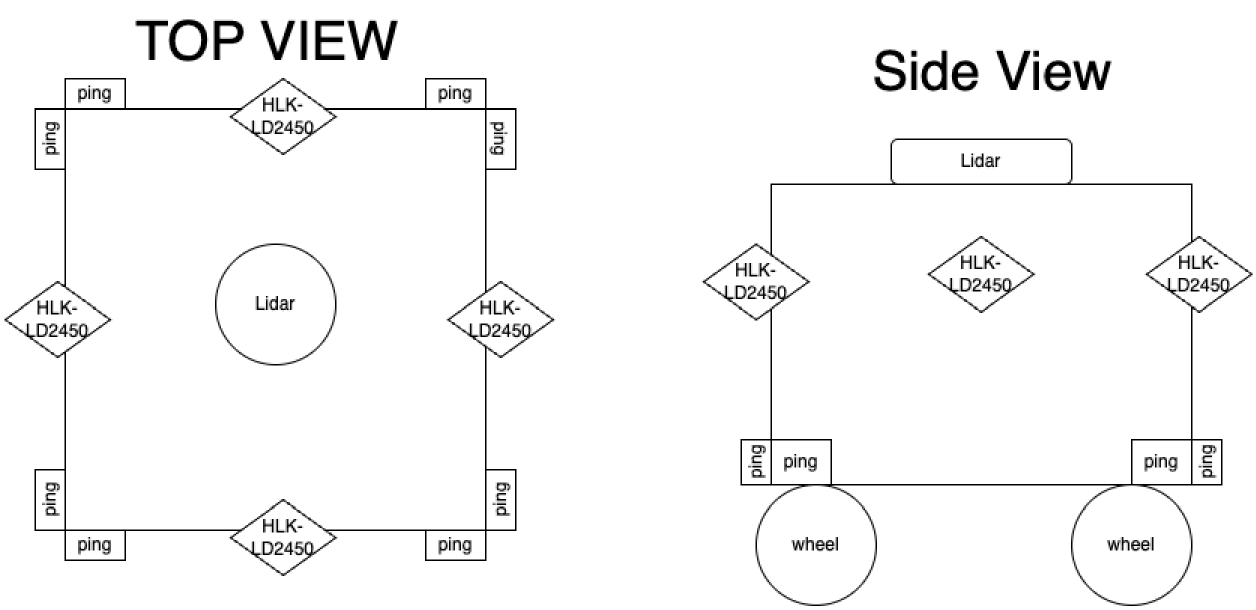
Over the summer, we have received all the sensors and parts needed for the robot. Other than that, we need to pick up some items like steel rope, screws, etc, from the hardware store. We have started assembling the Arm of the robot. We have been trying to run the GPS, which runs NMEA-0183.

One significant change of this project that was made is that instead of using the A121 radar sensor, we will use a combination of Lidar(LD14P), pings(Ultrasonic Distance Sensor), and HLK-LD2450(radar sensor mainly made for humans) sensors. We changed this because the radar was sensitive and noisier for our use. It's best used for distance measuring and not detecting objects' position, velocity. Still, it can be achieved using multiple sensors, but it will take longer to develop and is more expensive. Therefore, we are using those sensors. The Lidar should be able to detect most objects, but it can perform poorly in the rain and snow. So the ping sensor will help show the details that the Lidar couldn't pick up. Next, HLK-LD2450 can help confirm whether the object moving on the Lidar is a human, but we may not use this sensor if it is redundant or if we do not have enough time to implement it. So far, we have tested the lidar sensor, which can detect objects, and passed the experiment with the ping sensors.

<u>Testing the Lidar sensor</u>



<u>Arm Mechanism</u>



<u>The rough placement of the new sensors on the robot</u>

- **Past week accomplishments**

  - Sudipta Halder:Worked on making the arm mechanism, introduced and updated the team on the work done in the summer and discussed the change in sensors. Also introduced most of the sensors we got during summer.

  - Brodie Bates: This week started researching our new LiDAR Kit and got started on working with microcontrollers.

  - Daniel Pinilla: This week, focused on the machine learning part of the project and testing of the websocket communication

  - Om Shukla: The main purpose of this of this week for me was to remind myself on what the team accomplished last semester and what was accomplished last semester. Also started working on converting the HTML codebase to a REACT web app.

  - Mitchell Owen: The main purpose of this week was to go over what we accomplished last semester and catch up with the rest of the team. Haven't been working on the project this summer like some people did. Now that we are getting caught up we can all be on the same page and go attack this head on.

- **Pending issues**

  - Mitchell Owen: Making sure we all attend our meetings to get this done. We don't have weekly class time to keep us in order so it is on us to be able to meet and properly work on this.
  - Om Shukla: Im having trouble converting the HTML code to REACT app because of a library called Tailwind.

- **Individual contributions**

NAME	**Individual Contributions**	**Hours this week**	**HOURS cumulative**
Brodie Bates	Research new LiDAR Kit, Plan for building this week.	3	2
Om Shukla	Started work on the REACT app	2	2
Sudipta Halder	Worked on the arm	5	5
Daniel Mauricio Vergara	Worked on ML and Websocket testing	2	2
Mitchell Owen	Reviewing and game planning for the building stage.	2	2

- **<u>Plans for the upcoming week</u>**

    - Sudipta Halder: built the base of the robot with wood using tools at a workshop of student innovation . Attach the wheels and program the controllers. So we can sart

    - Brodie Bates: Meet with the team and begin the building process, along with getting code ready to be able to drive.

    - Om Shukla: Continue converting HTML code to REACT app code and then start implementing server integration into the website to display actual data.

    - Mitchell Owen: Meet with Brodie and Sudipta to at the minimum discuss how we can separate the hardware portion of this assignment and potentially start building our first model.

    - Daniel Mauricio Vergara: Work on the API connection for the Raspberry to the server and IOS app

- **<u>Summary of weekly advisor meeting</u>**

No Advisor meeting this week